

Overview

This document describes the binary format of BrickLink Stud.io **.conn** connectivity files. A .conn file contains a sequence of **Connectivity** records, each representing one connection point on a LEGO part. There is no global header or record count: records are read sequentially until end-of-file.

Key points:

- **Endianness:** All numeric values use little-endian (the default for .NET BinaryReader).
- **Record concatenation:** Records appear back-to-back without separators.
- **Structure:** Each record consists of a **Common Section** (type identifiers, transform matrix, local position) followed by a **Type-Specific Section**.

File Structure

1. **Records Only:** The file is a flat list of records.
2. **No Header:** There is no file-level metadata such as a record count.
3. **Little-Endian:** All Int16, Single, and Boolean values are little-endian.
4. **Sections per Record:**
 - **Common Section:** $2 \times \text{Int16} + 12 \times \text{Float32} = 56$ bytes
 - **Type-Specific Section:** size and layout vary by connection type

1. Common Section (56 bytes)

Every record begins with the following fields (offsets in bytes from record start):

Offset	Size	Type	Field	Description
0	2	Int16	TypeID	Connection type code (PhysicalT enum)
2	2	Int16	SubTypeID	Subtype or variant code
4	4	Float32	m00	Transform matrix element [0,0]
8	4	Float32	m10	Transform matrix element [1,0]
12	4	Float32	raw_m20	Raw [2,0] (negated when applied)

Offset	Size	Type	Field	Description
16	4	Float32	m01	Transform matrix element [0,1]
20	4	Float32	m11	Transform matrix element [1,1]
24	4	Float32	raw_m21	Raw [2,1] (negated when applied)
28	4	Float32	raw_m02	Raw [0,2] (negated when applied)
32	4	Float32	raw_m12	Raw [1,2] (negated when applied)
36	4	Float32	m22	Transform matrix element [2,2]
40	4	Float32	x	Local X coordinate
44	4	Float32	y	Local Y coordinate
48	4	Float32	raw_z	Raw Z coordinate (negated when applied)

Note: Negated fields must be inverted in code (e.g., `m20 = -raw_m20`). The 3×3 matrix combined with the vector (x, y, z) defines the part-relative orientation and position of the connector.

2. Type-Specific Sections

Each record's payload depends on its TypeID. The following sections describe the layout for each supported type.

These are the Ids:

Stud	0
Hole	1
Axle	2
Ball	3
Fixed	4
Hinge	5
Rail	6
Slider	7

2.1 Stud

Defines a male stud connection and its grid of contact points.

Field	Type	Description
height	Int16	Number of grid rows minus 1
width	Int16	Number of grid columns minus 1
matrix	Int16[height+1][width+1] ×2	For each cell:

- Altitude (Int16): height in LDU units
- OccupiedArea (Int16): nonzero if that cell is active in the stud footprint

2.2 Hole

Female counterpart to Stud. Layout is identical:

- height (Int16)
- width (Int16)
- matrix of (Altitude, OccupiedArea) pairs

2.3 Axle

Represents a Technic axle or axle hole.

Field	Type	Description
isStartCapped	Boolean	true if start end is closed
isEndCapped	Boolean	true if end end is closed
length	Float32	Axle length in LDU units
isGrabbing	Boolean	true if this axle grabs into a hole
isRequireGrabbing	Boolean	true if hole must grab to form a connection

2.4 Ball Joint

Defines a ball-and-socket joint.

Field	Type	Description
radius	Float32	Ball radius in LDU units
socketDepth	Float32	Depth of the cup-shaped socket
rangeOfMotion	Float32	Maximum rotation angle (degrees or radians, depending on implementation)

Parsing pattern: after common section, read three Single values in order.

2.5 Fixed

A rigid connector without movement.

No additional fields. The record ends immediately after the common section.

2.6 Hinge

Represents a hinged joint with rotation limits.

Field	Type	Description
minAngle	Float32	Minimum rotation angle
maxAngle	Float32	Maximum rotation angle
axisDirection	Int16[3]	Optional vector indices defining hinge axis (if present)

Parsing: read two Single values, then optionally three Int16 values if defined.

2.7 Rail

Defines a sliding connection along a rail or track.

Field	Type	Description
length	Float32	Track length in LDU units

Field	Type	Description
isRetractable	Boolean	true if the slider can be retracted/folded

Read: one Single, then one Boolean.

2.8 Slider

A linear slider connector.

Field	Type	Description
isDamped	Boolean	true if movement is damped
range	Float32	Maximum slide distance in LDU units

Read: one Boolean, then one Single.

3. Example Byte Map

Below is an example for a Stud record with height = 1, width = 1 (2×2 grid):

00-01: TypeID = Stud

02-03: SubTypeID

04-07: m00

08-11: m10

12-15: raw_m20

16-19: m01

20-23: m11

24-27: raw_m21

28-31: raw_m02

32-35: raw_m12

36-39: m22

40-43: x

44-47: y

48-51: raw_z

52-53: height = 1

54-55: width = 1

56-57: Altitude(0,0)

58-59: Occupied(0,0)

60-61: Altitude(0,1)

62-63: Occupied(0,1)

64-65: Altitude(1,0)

66-67: Occupied(1,0)

68-69: Altitude(1,1)

70-71: Occupied(1,1)

4. Summary

- A .conn file is a sequence of binary records, each starting with a **Common Section** (TypeID, SubTypeID, 3×3 matrix, XYZ) followed by type-specific fields.
- All numeric fields use little-endian encoding.
- Stud/Hole records include a grid of contact heights and occupancy flags.
- Axle, Ball, Hinge, Rail, Slider each have dedicated parameters that control mechanical behaviour in Stud.io.
- You can implement a parser in any environment by following these byte offsets and field types.